Madhusudan Bhukta

Backend Engineer | Node.js & AI Systems Architecture

Email | GitHub | LinkedIn | Portfolio | Bhubaneswar, India

SUMMARY

Backend engineer with hands-on production experience building REST APIs, integrating AI/ML services into Node.js systems, and optimizing database and caching layers. Currently shipping features in a live CRM/HRM platform serving hundreds of active users. Deployed AI services to AWS using Docker, custom domains, and SSL certificates.

SKILLS

Languages: JavaScript (ES6+), Python, SQL
Backend: Node.js, Express.js, FastAPI, REST API Design
Databases: MongoDB, Redis, ChromaDB
Architecture: Microservices Communication, RAG Pipelines, Hybrid Search, Vector Databases
Auth & Performance: JWT, RBAC, Refresh Tokens, Redis Caching, Rate Limiting, Indexing, Pagination
Cloud & DevOps: AWS (EC2, Security Groups), Docker, Docker Compose, Nginx, SSL/Certbot, Linux, Git, Render, Vercel

EXPERIENCE

Backend Developer Intern | *Detagenix Pvt. Ltd. (Remote)* **May 2026–Aug 2026**

- Built and shipped 26+ REST API endpoints in a production CRM/HRM system, bridging Node.js/Express with Python FastAPI AI services for cross-service data flow.
- Resolved 3+ critical production bugs across frontend-backend integration, improving API reliability and reducing error rates in live user workflows.
- Designed API contracts between Node.js and FastAPI services, enabling real-time risk anomaly detection and automated report generation with sub-180ms latency.

PROJECTS

Project KYH — AI Fitness Agent *Python, LangGraph, Node.js, Docker Compose, AWS EC2, MongoDB*
Live System | Source Code

- Architected an asynchronous multi-agent health coach utilizing **LangGraph**, orchestrating 7 discrete graph nodes and conditional subgraphs to decouple intake tracking from macro planning logic.
- Containerized and orchestrated split Node.js and Python microservices alongside persistent state layers via **Docker Compose**, deploying the multi-runtime system onto a dedicated **AWS EC2** cloud instance.
- Implemented multi-day user session persistence using LangGraph checkpointing synced with MongoDB Atlas to track and evaluate dietary behavior trends across a 30-day window.

ShopNexus — E-Commerce Backend *Node.js, Express, MongoDB, Redis, Stripe*
Live Demo | Source Code

- Profiled live deployment thresholds under micro-burst concurrency using **k6**, pushing **3,100+ requests** at a peak of **50 concurrent virtual users** to systematically locate database bottlenecks and infrastructure compute constraints.
- Built complete Stripe payment lifecycle (checkout, webhooks, refunds) with idempotency handling and 12+ domain-driven MongoDB models.
- Implemented JWT auth with refresh token rotation, RBAC middleware, and rate limiting to protect 25+ API routes.

RAG-DocumentQnA — AI Document Search *Node.js, Python, ChromaDB, LangChain*
Source Code

- Built dual-service architecture (Node.js API gateway + Python AI backend) supporting PDF, TXT, and DOCX parsing with per-user session isolation.
- Reduced LLM hallucinations by implementing Hybrid Search (Dense embeddings + BM25) with a Cross-Encoder re-ranker, improving retrieval precision across 3 document types.

EDUCATION

Bachelor of Computer Applications (BCA) **2024–2027**
NIIS Institute of Business Administration, Bhubaneswar, Odisha | **CGPA: 8.1 (1st Sem)**